

Week 10: Working with an image or document

Generative AI | MSc teaching notes and further reading | 2027

Learning target

Distinguish reading from independently checking a document value.

Context and teaching plan

A synthetic invoice image has fields supplied by a prepared transcription.

Contact time	Suggested use
Two hours: concepts	Case introduction, four ideas, worked example and misconception.
Two hours: practice	10 min recall, 20 min demo, 40 min guided practice, 25 min independent change, 15 min review, 10 min exit check.

Explain the key ideas

Multimodal input

A system may process text, images or other signals. Here the transcription stands in for an image reader.

Teacher question: Where does multimodal input appear in the weekly case, and what could go wrong if we omit it?

Extraction

Read fields present in the document and retain a pointer to the source.

Teacher question: Where does extraction appear in the weekly case, and what could go wrong if we omit it?

Verification

Recalculate totals and compare against the listed values.

Teacher question: Where does verification appear in the weekly case, and what could go wrong if we omit it?

Uncertainty

Mark unreadable or absent fields for review rather than inventing a value.

Teacher question: Where does uncertainty appear in the weekly case, and what could go wrong if we omit it?

Worked example and answer guide

- Input: Invoice shows 2 items at €20 plus €5 tax.
- Change or question: Calculated total is €45.
- Supported observation: Listed total of €54 triggers review, even if transcription is syntactically valid.

Reasoning: state the given evidence first, then the calculation or rule, then the conclusion. If a value is absent, say what cannot be concluded. Keep any simulated mechanism distinct from a trained model or deployed service.

Teach the week as a story

The €54 invoice: An invoice image lists two items at €20 and €5 tax, yet displays €54 total. A model reads €54 correctly. Should the payment pass?

Ask for a quick prediction before each technical explanation. Return to the prediction after the example and ask what evidence changed the answer. The analogies below are memory aids, and each has a stated limit.

Multimodal input: analogy, mechanism and check

Familiar picture: A person can read a photo and a text note together.

Familiar part	Technical meaning
Photo	visual input
Reader	OCR or vision step
Typed note	transcribed text

Where this analogy breaks: This notebook uses a prepared transcription; it does not run OCR or test a vision model.

Actual idea: A system may process text, images or other signals. Here the transcription stands in for an image reader.

Work it aloud: The prepared text records quantity 2, unit price €20, tax €5 and listed total €54.

Ask before revealing: What capability is represented by the supplied transcription?

Supported response: An upstream reading step, whose accuracy still needs checking.

Look for this misconception: The notebook itself recognises image pixels.

Extraction: analogy, mechanism and check

Familiar picture: An auditor writes the page and line beside every copied number.

Familiar part	Technical meaning
Printed number	visible field
Copied value	extraction
Page pointer	source location

Where this analogy breaks: A pointer can be wrong or ambiguous, especially with poor image quality.

Actual idea: Read fields present in the document and retain a pointer to the source.

Work it aloud: Record “listed_total = €54” with the invoice field location; keep calculation separate.

Ask before revealing: What does a source pointer let another person do?

Supported response: Find and verify the value in the original document.

Look for this misconception: A correctly formatted field proves its source.

Verification: analogy, mechanism and check

Familiar picture: A cashier recomputes a bill instead of trusting the printed sum.

Familiar part	Technical meaning
Items	$2 \times \text{€}20$
Tax	€5
Independent sum	€45

Where this analogy breaks: Real invoices may have discounts and other rules; specify the case assumptions.

Actual idea: Recalculate totals and compare against the listed values.

Work it aloud: $2 \times \text{€}20 + \text{€}5 = \text{€}45$; listed total €54, a €9 discrepancy.

Ask before revealing: Does reading €54 correctly make it a correct payable total?

Supported response: No. The independent arithmetic disagrees by €9.

Look for this misconception: Accurate OCR implies accurate arithmetic.

Uncertainty: analogy, mechanism and check

Familiar picture: A reviewer marks an unreadable receipt field with a question mark.

Familiar part	Technical meaning
Blurred area	uncertain observation
Question mark	review flag
Second look	verification

Where this analogy breaks: Uncertainty needs a clear workflow; marking everything uncertain is unhelpful.

Actual idea: Mark unreadable or absent fields for review rather than inventing a value.

Work it aloud: If INV-08/INV-03 cannot be distinguished, store an uncertain ID and request review.

Ask before revealing: What would be unsafe to store?

Supported response: A guessed invoice ID presented as certain.

Look for this misconception: A plausible guess should be saved silently.

New situation for discussion

A blurred invoice number looks like INV-08 or INV-03. Which field should be marked uncertain?

Memory cue: Read the source, preserve field pointers, calculate independently, and flag discrepancies.

Guided laboratory

Check two synthetic invoices, including a mismatched total.

1. Inspect a synthetic source transcription
2. Extract amount, quantity, tax and stated total
3. Calculate the expected total independently
4. Flag mismatches and retain the source ID

Individual change: Correct the listed total and rerun the checker. Expected evidence: Source-linked field table and calculation.

Checks and common error

Measure: Number of correctly checked fields / fields reviewed. Record the input, setting, source version, result and the relevant numerator and denominator.

Plausible incorrect claim: The notebook performs OCR on a real image.

Correction: This core lab uses prepared transcription to teach verification. Image/OCR integration is an optional extension.

If learners disagree, reproduce the same case and compare the underlying source or expected answer. Record unresolved cases for instructor review.

Further discussion

- How could the conclusion change if the source or input changes?
- Which part of the result is observed and which part is an assumption?
- What simple test would detect the failure described above?

References and optional extension

Core readings are official vendor documentation or original research. Read the relevant section; an account or paid API is unnecessary for this notebook. Page details may change before 2027.

Google, Develop a generative AI application.

<https://docs.cloud.google.com/docs/ai-ml/generative-ai/develop-generative-ai-application>

OpenAI, Evaluation best practices. <https://developers.openai.com/api/docs/guides/evaluation-best-practices>

Optional extension: replace one classroom proxy with a locally approved model or service, keep the original baseline, repeat the same checks, and state the additional privacy and cost assumptions.

Curriculum scope: the supplied 12-week teaching sequence, recorded in docs/SYLLABUS_ALIGNMENT.md. The original module descriptor is retained in sources/originals/07-Descriptor-Generative-AI-new-.docx. Supporting project notes: Deep-research.txt, RAG.txt, AI-Agents.txt, MCP-&-Vector-Database.txt, Multi-Agent-AI-System.txt, Eval-&-Fine-tune-&-Loop-Engineering.txt. These are planning sources, not empirical performance evidence.

Student revision and exam preparation

Outcomes: LO2, LO4 | 2 hours concepts + 2 hours guided practice

The idea in plain language

A multimodal system can accept an image and text. Reading a value from an image and verifying that value are different operations. Retain the original source location, mark unreadable fields and recalculate totals with ordinary code.

A worked example

Two items at 20 plus tax 5 imply 45. If the image says 54, a perfect transcription of 54 is still inconsistent with arithmetic. Flag the mismatch and ask for review; the discrepancy alone does not prove fraud.

Practice before reading the answers

1. What is the difference between extraction and verification? (4 suggested practice marks)

2. Calculate the total for 2×20 plus 5 tax. (4 suggested practice marks)

3. How should an unreadable digit be handled? (4 suggested practice marks)

Draw the main diagram from memory, explain each arrow, then check it against the slides. Answer the three questions without notes before reading the model answers.

Model answers and marking guidance

1. Extraction records the visible value. Verification checks it against arithmetic, evidence or rules.

2. 45. A listed value of 54 differs by 9 and should be reviewed.

3. Mark it uncertain or missing and request a clearer source; do not silently guess.

For each four-mark practice answer: two marks for the correct idea, one for a relevant example or calculation, one for an appropriate explanation or limitation. These suggested practice marks do not change the module assessment scheme.

Official reading and video support

G7 Google Developers: Image understanding

<https://ai.google.dev/gemini-api/docs/image-understanding>

Study focus: Multimodal image inputs. The core notebook uses a prepared transcription.

O1 OpenAI: Evaluation best practices

<https://developers.openai.com/api/docs/guides/evaluation-best-practices>

Study focus: Task-specific testing, repeated outputs and separate evaluation cases.

Selected source concepts inform the lesson. Examples, figures, questions and solutions are original classroom material; no exam questions are claimed to be official.